

Domovoy Crater (Ariel) 203
Doppler effect 35
Doppler spectroscopy 297, 297
Dorado (the Goldfish) **421**
Beta (β) Doradus 421
see also Large Magellanic Cloud
R Doradus 421
see also Tarantula Nebula (30 Doradus)
double binary stars 274
Double Cluster 370, 370, 496
double-slit test 34
double stars *see* binary stars
Dove *see* Columba
DR 6 **243**
Draco (the Dragon) **355**, 460, 466
Abell 2218 **334–335**
Cat's Eye Nebula **258**, 355, 355
Eltanin (Gamma (γ) Draconis) 355
Mu (μ) Draconis 355
Nu (ν) Draconis 355
Omicron (ο) Draconis 355
Psi (ψ) Draconis 355
Spindle Galaxy **317**
16 Draconis 355
17 Draconis 355
39 Draconis 355
40 Draconis 355
41 Draconis 355
Dragon *see* Draco
Drake, Frank 57
Draper, Henry 241
Dreyer, J. L. E. **239**
Dubhe (Alpha (α) Ursae Majoris) 72, 360
Hertzsprung–Russell (H–R) diagram 232
Duck Bay (Mars) 164
Dumbbell Nebula (M27) 89, 257, 384, 384, 472, 473
Dunlop, James 260
Dürer, Albrecht 347
dust
interstellar medium 24, 228
storms on Mars 159
zodiacal light 75, 75
dwarf elliptical galaxies 16–17, 304, 304, 310
Canis Major Dwarf 310
galaxy clusters 326, 326
SagDEG **310**
dwarf planets *see* Kuiper Belt Objects
dwarf stars
black dwarfs 235, 237, 266
brown dwarfs 25, 25, 234
red dwarfs 25, 25, 235, 299
see also white dwarfs
Dysnomia 212, 212

E

e Puppis 409
Eagle *see* Aquila
Eagle Crater (Mars) 163
Eagle Nebula (IC 4703) 238, **244–245**, 380, 380, 467
Earth 8, 25, **126–135**, 142–143
asteroids 170
atmosphere and weather 128, 128
aurorae 74, 107
axis of rotation 64, 64

Earth *cont.*
celestial sphere **62–63**
climate 126
Earthrise 142–143
eclipses **67**
features formed by water **134–135**
geological features **130–133**
life 56–57, 129, 129
lights in the sky **74–75**
magnetic field 127, 127
meteorite craters 221, 221, **222–223**
meteorites 220
the Moon 136, 137, 138, 138
orbit and spin 102, 126, 126
plate tectonics 128, 128
seasons 65, 65, 126
size 22
structure 126, 126
surface features 129
earthquakes
Mercury 112
meteorite impacts 221
eclipses **67**
eclipsing binary stars 274, 274, 370
Alpha (α) Herculis (Ras Algethi) **285**
Eta (η) Geminorum (Propus) **284**
ecliptic 64, 65, 126
Eddington, Sir Arthur **251**
Edgeworth, Kenneth 211
Edgeworth–Kuiper Belt 211
Egg Nebula (CRL 2688) **258**
Egypt, constellations 346
Eight–Burst Nebula (NGC 3132) 254–255, 410, 443
Einstein, Albert 31, **40**
cosmological constant 58
energy and mass 41
general theory of relativity **42–43**, 51
mass and energy 58
Mercury's orbit 110
principle of equivalence 42, 42
special theory of relativity **40–41**
Eirene 181
Eistla Regio (Venus) **121**
Elara 180
Electra, in mythology 373
electromagnetic (EM) force 30, 30, 49
electromagnetic (EM) radiation **34–37**
“false color” images 37, 37
observing **34–37**
Sun 104
electron degeneracy pressure, white dwarfs 266
electrons 28, 28–29
Big Bang 49, 50, 50–51
Big Chill 59
chemical elements 29
forces 30, 30
molecules 29
photoelectric effect 34, 34
plasma 30
synchrotron mechanism 320
electroweak epoch 48, 48–49
electroweak force 49
elementary particles 31, 48, 50
elements 29
formation of 55, 266, 266
high-mass stars 236
planet formation 235

elements *cont.*
spectroscopy 35, 35
star formation 234
supergiant stars 254
Elephant's Trunk Nebula 243, 243
Elias 2–27 296
ellipses, orbits 39, 39
elliptical galaxies 26, **304**
classification 302, 302
distribution 306
ESO 325–G004 304
galaxy clusters 326, 327
M60 **317**
SagDEG **310**
Elnath (Beta (β) Tauri) 232, 359, 372
ELODIE 297
ELT (Extremely Large Telescope) 37
Eltanin (Gamma (γ) Draconis) 355
Elysium Planitia (Mars) **162**
emission nebulae 24, 35, 228, 240
Carina Nebula **247**, 248–249
DR 6 **243**
Eagle Nebula **244–245**
IC 1396 **243**
IC 2944 **246**
Lagoon Nebula **243**
M43 241, 391
NGC 604 311, 311
NGC 2359 264
Omega Nebula 240, 400, 401
Orion Nebula **241**
RCW 49 **247**
Seagull Nebula 238
Sharpless 29 (SH 2–29) 246
Trifid Nebula **246**
emission spectrum 35, 35, 233
planetary nebulae 255
Wolf–Rayet stars 255
Enceladus 190, 191, **194**
Encke, Comet 214, **217**
Encke, Johann 217
Encke gap, Saturn's rings 191
Encounter 2001 message 57
end points, stellar **266–273**
Endurance Crater (Mars) **166–167**
energy
atomic bomb 41
atoms 28
Big Bang 48
convection 250
dark energy 27, 54, 58–59, 58, 339
electromagnetic (EM) radiation 34, 34
fate of universe 58–59
ionization 28
luminosity 233
main-sequence stars 250
mass 41, 41
nuclear fission and fusion 31, 31
photons 34
protostars 239
radiation 250
rotation 39
Saturn 189
stars 232
states of matter 30
strong nuclear force 30
Sun 104
supernovae 266
Enif (Epsilon (ε) Pegasi) 386, 478
Enki Catena (Ganymede) 215

Ensisheim meteorite **222**
Eos Chasma (Mars) 159
Epimetheus 190, **192**
EPOCh mission 219
EPOXI mission 219
Epsilon (ε) Aurigae (Almaaz) **281**, 283, 283, 359
Epsilon (ε) Bootis (Izar) **277**, 363, 363, 460
Epsilon (ε) Carinae 411
Epsilon (ε) Cepheus 356
Epsilon (ε) Corvi 397
Epsilon (ε) Herculis 364
Epsilon (ε) Hydrae 394
Epsilon (ε) Indi 416
Epsilon (ε) Lupi 399
Epsilon (ε) Lyrae **276**, 365
Epsilon (ε) Normae 414
Epsilon (ε) Orionis (Alnilam) 232
Epsilon (ε) Pegasi (Enif) 386, 478
Epsilon (ε) Sagittarii 400
Epsilon (ε) Sculptoris 404
Epsilon (ε) Ursae Majoris (Alioth) 72, 360
Epsilon (ε) Volantis 422
equator, celestial sphere 62, 63
equatorial mountings, telescopes 83, 83, 84, 86–87
equatorial sky charts 350–353
equinoxes 65, 65, 126
Pisces 388
precession 371
sky guide 442
Equuleus (the Foal) **385**
Gamma (γ) Equulei 385
1 Equulei 385
Erichthonius 359
Eridanus (the River) **406**, 485, 497
NGC 1531 301
NGC 1532 301
see also Achernar (Alpha (α) Eridani)
Omicron (ο) Eridani **276**, 406, 406
Theta (θ) Eridani 406
32 Eridani 406
40 Eridani B 232
Erie, Lake (Earth) 134
Erinome 181
Eris 210, 212, 212
Eros 13, 170, 172, **176–177**
mythology 388
erosion
Mars 164, 164
Venus 119
Erriapus 191
Ersa 180
eruptive variable stars 282–283
U Geminorum **284**
ESA *see* European Space Agency
Eskimo Nebula (NGC 2392) 374, 374, **259**
ESO 97–G13 (Circinus Galaxy) **322**
ESO 350–G40 (Cartwheel Galaxy) 319
ESO 510–G13 **318**
Eta Aquarii meteor shower 387, 454, 455
Eta (η) Aquarii 387, 455
Eta (η) Aquilae **286**, 383
Eta (η) Carina Nebula *see* Carina Nebula
Eta (η) Carinae 247, 248–249, 256, **262**, 411, 411, 443, 449

Eta (η) Cassiopeiae 357
Eta (η) Geminorum (Propus) **284**, 374
Eta (η) Herculis 364
Eta (η) Lupi 399
Eta (η) Piscium 388
Eta (η) Tauri (Alcyone) **277**, 291, 372
Eta (η) Ursae Majoris (Alkaid) 72, 360
Eta (η) Ursae Minoris 354
ethane
Jupiter 180
Saturn 189
“ether” 40
Euanthe 181
Eudoxus 346
Eukelade 180, 181
Eupheme 181
Euporie 181
Europa 25, 180, **182–182**
possibility of life 57
European Center for Nuclear Research (CERN) 49, 32–33
European Space Agency (ESA)
BepiColombo space probe 113, 113, 115
CHEOPS (Characterizing Exoplanet Satellite) 299
Giotto mission 217
Hipparcos satellite 70, 70
Planck Observatory 36, 54
Rosetta mission 172, 218–219
Solar Orbiter (solO) 105, 105
Eurydome 181
evaporating gaseous globules (EGGs) 238, 244
event horizon, black holes 43, 267
Event Horizon Telescope 26, 323
Everest, Mount (Earth) 132, 132
evolution
galaxies **306–309**
galaxy clusters 327
life 129
multiple stars 274, 274
star clusters 289
stars **234–237**
ExoMars Trace Gas Orbiters 152
exoplanets *see* extrasolar planets
exotic particles 31, 48
expanding space **44–45**, 58, 335, 338–339, 339
Explorer 1 satellite 127
extrasolar planets **296–299**
extraterrestrial life 57
Extreme Deep Field 94
Extremely Large Telescope (ELT) 37
extremophile organisms 57
eyepieces, telescopes 82, 83, 85, 85
eyes, adjusting to dark 76

F

f number 83
F stars 233
Fabricius, David 285
faculae, Sun 85, 106, 108–109